



Laboratory Report # 3

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Date Completed: September 20, 2024

Laboratory Exercise Title: Structural Modeling of Combinatorial Circuits

Target Course Outcomes:

CO1: Create descriptions of digital hardware components such as in combinational and sequential circuits using synthesizable Verilog HDL constructs.

CO2: Verify the functionality of HDL-based components through design verification tools.

Exercise 3A: 2x4 Decoder. Under a new project, create a Verilog HDL description of a 2-to-line decoder with active-high enable (E) signal using gate primitives and structural modeling. For structural modeling, derive the Boolean functions and draw the logic diagram of the circuit. Synthesize the design.

INPUTS			OUTPUT			
E	A1	A0	D			
			D3	D2	D1	D0
0	x	x	0	0	0	0
1	0	0	0	0	0	1
	0	1	0	0	1	0
	1	0	0	1	0	0
	1	1	1	0	0	0

Table 1. 2x4 Decoder Circuit Truth table

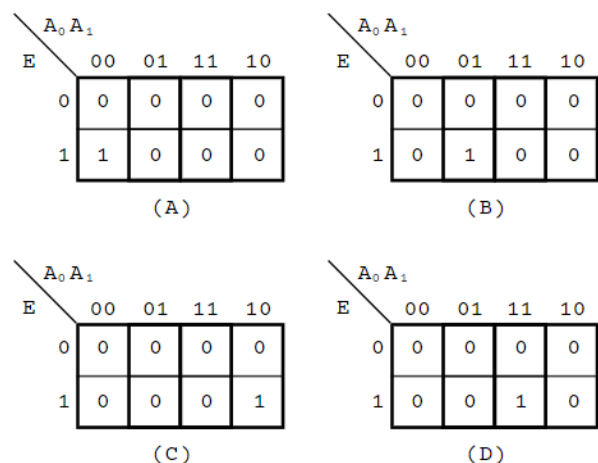


Figure 1. K-Map of the 2x4 Decoder Outputs. (A) D0 output. (B) D1 Output. (C) D2 Output. (D) D3 Output.

Based on the K-maps generated from Table 1, boolean functions can be made to represent the four outputs of the decoder:

$$D0 = E * (A0)' * (A1)'$$

$$D1 = E * A0 * (A1)'$$

$$D2 = E * (A0)' * A1$$

$$D3 = E * A0 * A1$$

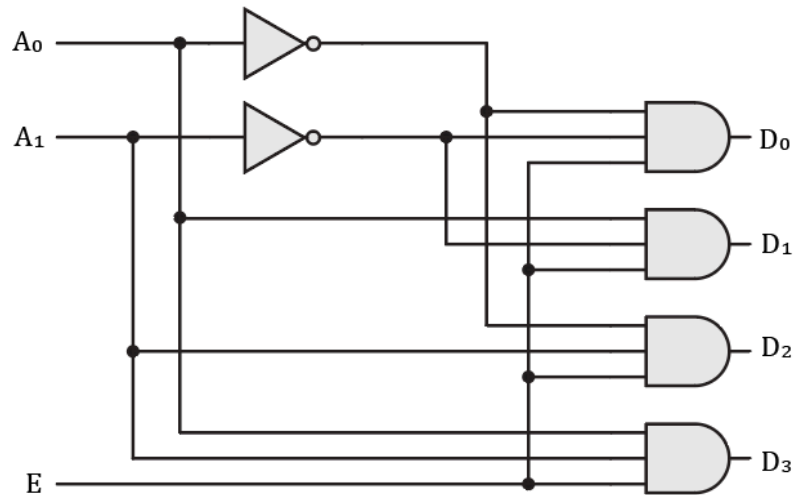


Figure 2. Logic Diagram of a 2x4 Decoder Circuit.

Using the equations that are derived from the K-maps, the gate primitives in Verilog can be utilized to simulate the logic behind the outputs.

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/*****
* AUTHOR:      CYRIL ANDRE B. DURANGO
* CLASS:       CPE 3101L - INTRODUCTION TO HDL
* SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
* -----
* PROJECT FILE TITLE: Decoder_2by4.v
* PROJECT DESCRIPTION:
*   A Verilog File of a 2x4 Decoder Circuit
*
*****/

module Decoder_2by4(A,E,D);
    input  [1:0] A;
    input      E;
    output [3:0] D;
    wire    [1:0] notA;

    not    notA0(notA[0],A[0]);
    not    notA1(notA[1],A[1]);

    and    D0(D[0], E, notA[0], notA[1]);
    and    D1(D[1], E, A[0], notA[1]);
    and    D2(D[2], E, notA[0], A[1]);
    and    D3(D[3], E, A[0], A[1]);

endmodule

```

Figure 3. Verilog HDL File simulating a 2x4 Decoder Circuit

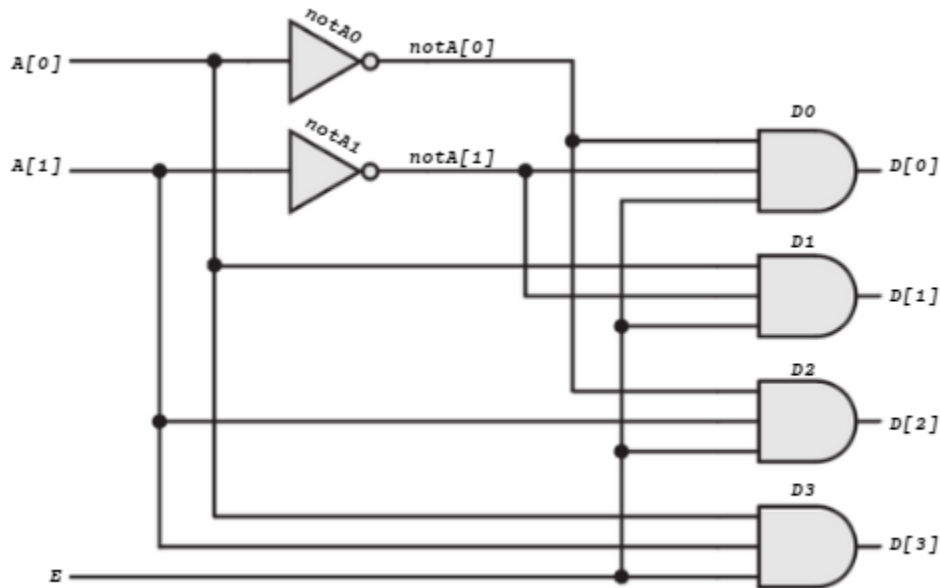


Figure 4. Logic Diagram of a 2x4 Decoder Circuit with annotated I/O ports and internal signals.

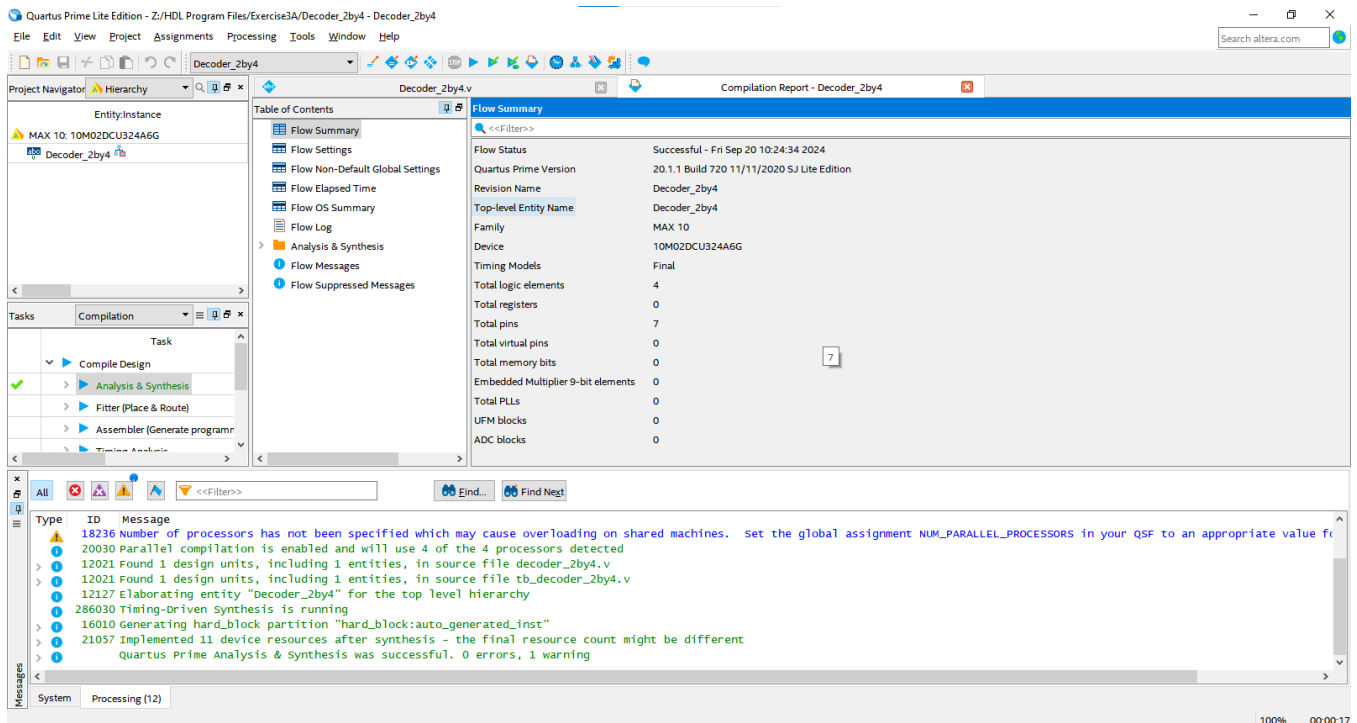


Figure 5. 2x4 Decoder Circuit Flow Summary Compilation

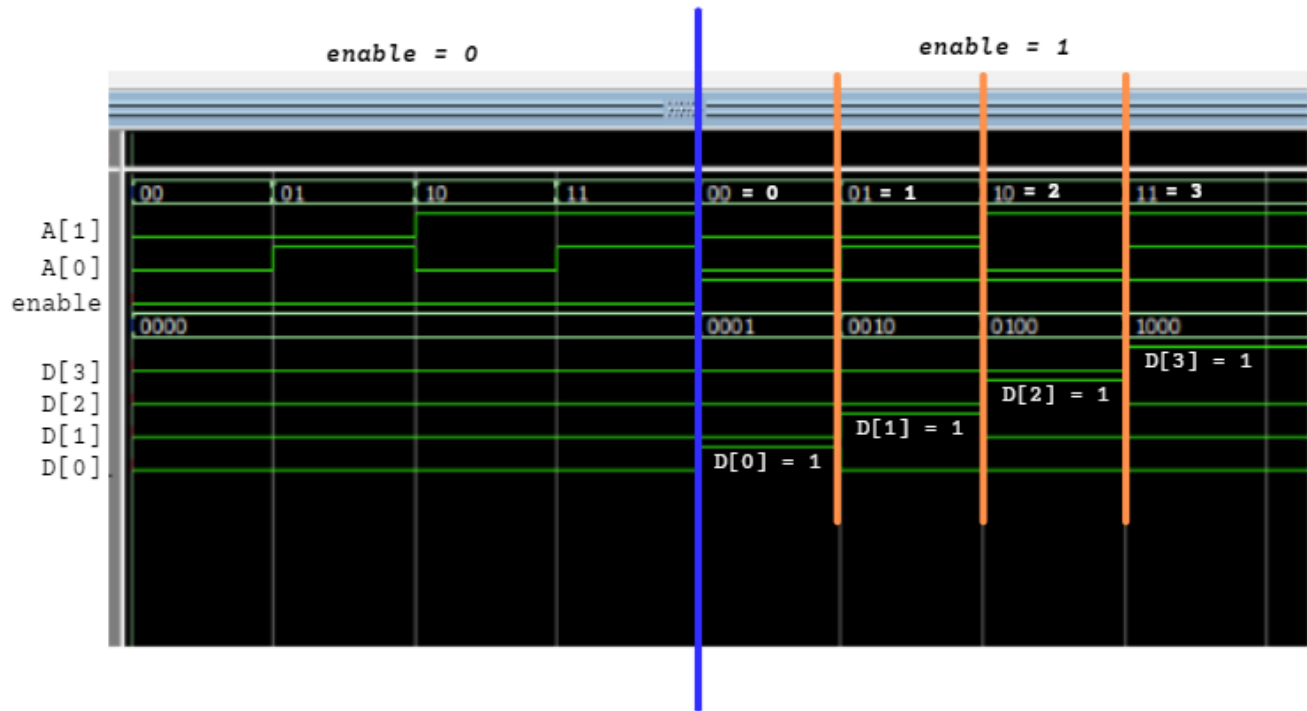


Figure 6. 2x4 Decoder Generated Waveform

The basic concept of a decoder is to generate an output given a binary input. A 2x4 Decoder requires 2 binary inputs, which could generate four unique values (00, 01, 10, 11). These four unique values have their corresponding output.



Exercise 3B: 3x8 Decoder. Using the 2x4 decoder in Exercise 3A, derive and create a structural description of a 3-to-8-line decoder with active-high enable (E).

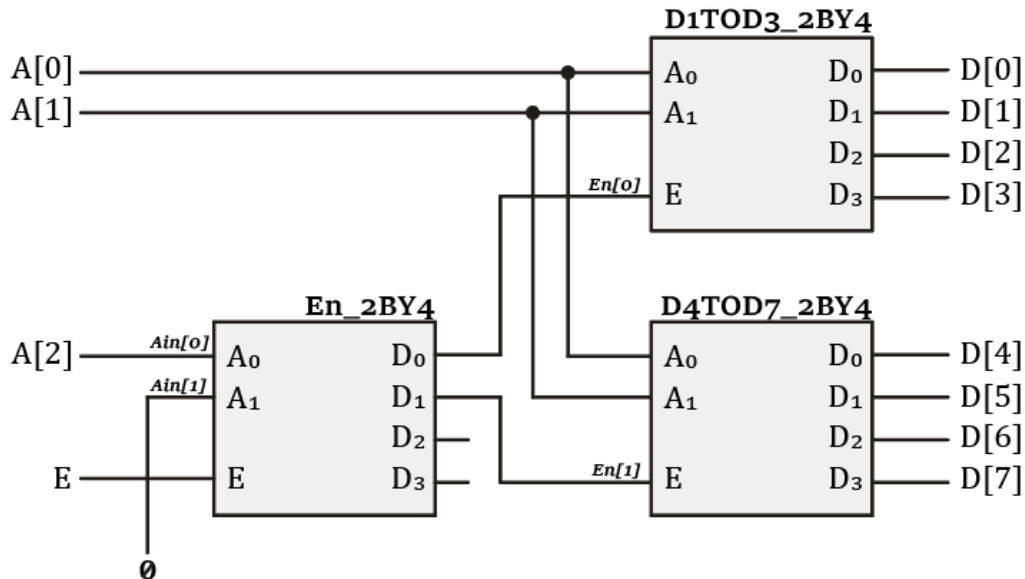


Figure 7. Logic Diagram of a 3x8 Decoder Circuit using 2x4 Decoders with annotated I/O ports and internal signals.

```
module Decoder_3by8(A,E,D);
    input [2:0] A;
    input E;
    output [7:0] D;
    wire [1:0] Ain;
    wire [3:0] En;
    wire nA2;

    and A3(Ain[1], 0, A[2]);
    and A2(Ain[0], 1, A[2]);

    Decoder_2by4 En_2by4 ( .A (Ain[1:0]) ,
                          .E (E),
                          .D (En[3:0])
    );

    Decoder_2by4 D1TOD3_2by4 ( .A (A[1:0]) ,
                              .E (En[0]),
                              .D (D[3:0])
    );

    Decoder_2by4 D4TOD7_2by4 ( .A (A[1:0]) ,
                              .E (En[1]),
                              .D (D[7:4])
    );

endmodule
```

Figure 8. Verilog HDL File simulating a 3x8 Decoder Circuit

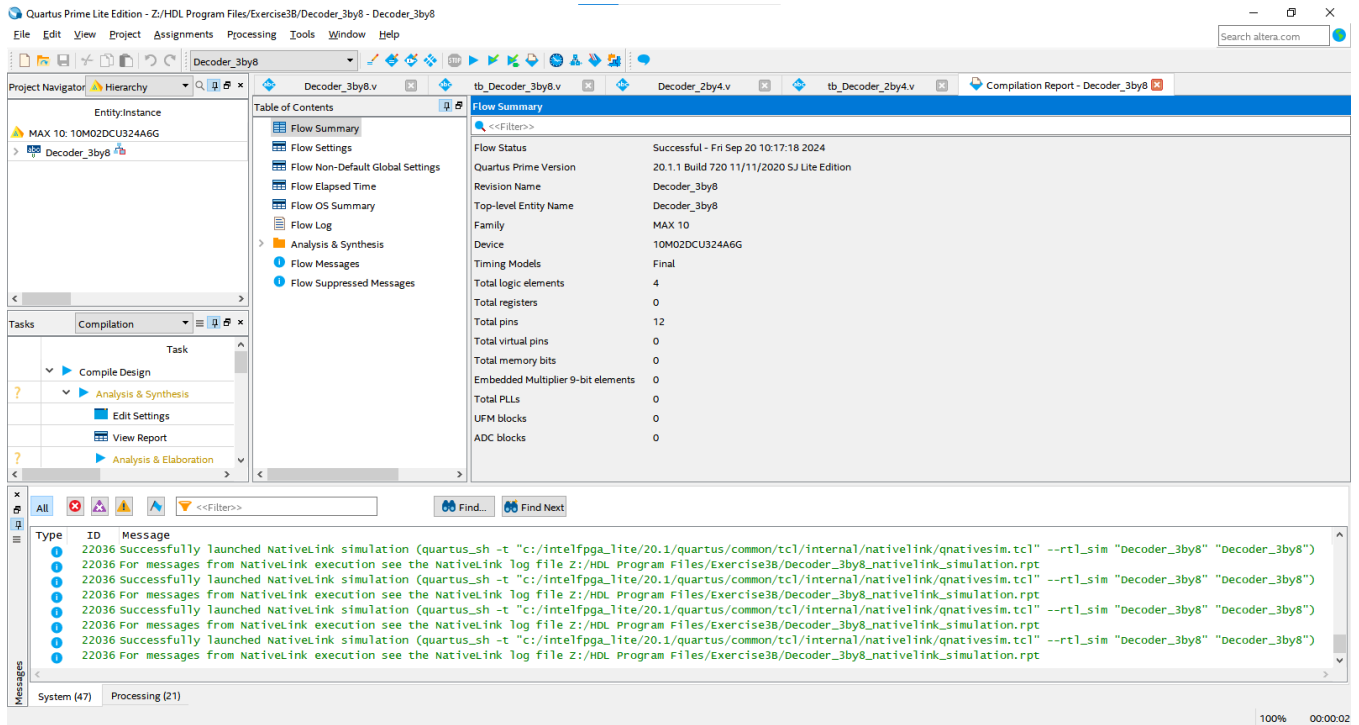


Figure 9. 3x8 Decoder Circuit Flow Summary Compilation

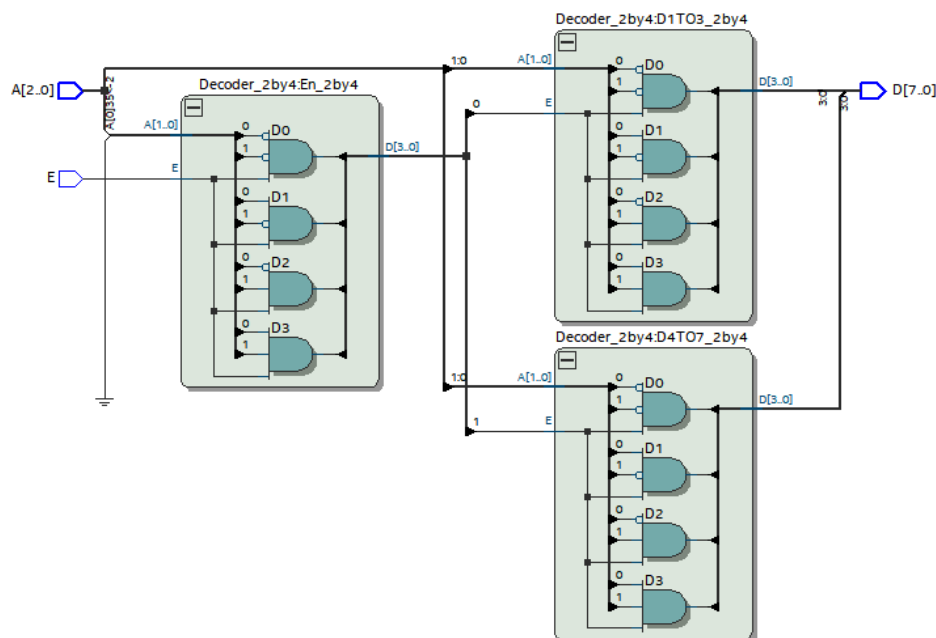


Figure 10. RTL Viewer of the simulated 3x8 Decoder Circuit

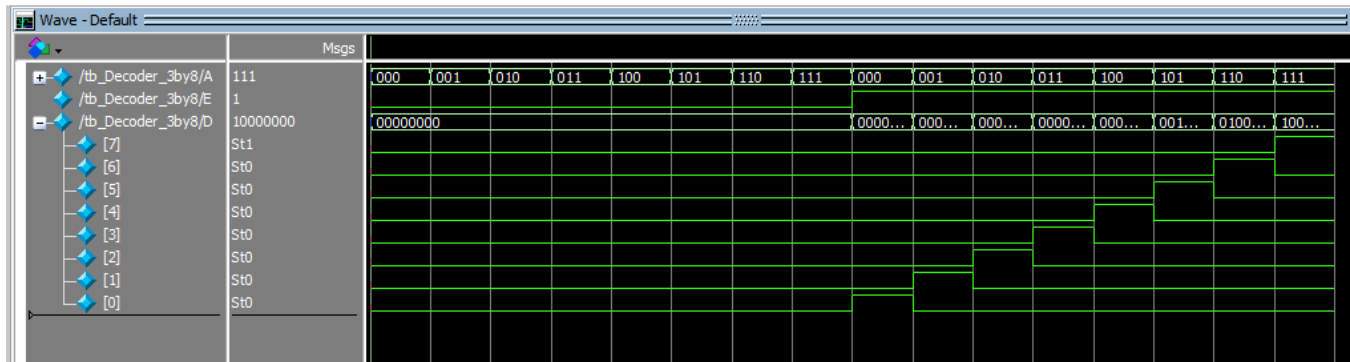


Figure 11. 3x8 Decoder Waveform

A similar output is to be observed in the waveform of the 3x8 Decoder Waveform to the 2x4 Decoder Waveform (see Figure 6). When the enable is 0, the output is at logic-0. When the enable is 1, it can be seen that each input combination generates a unique logic-1 output.